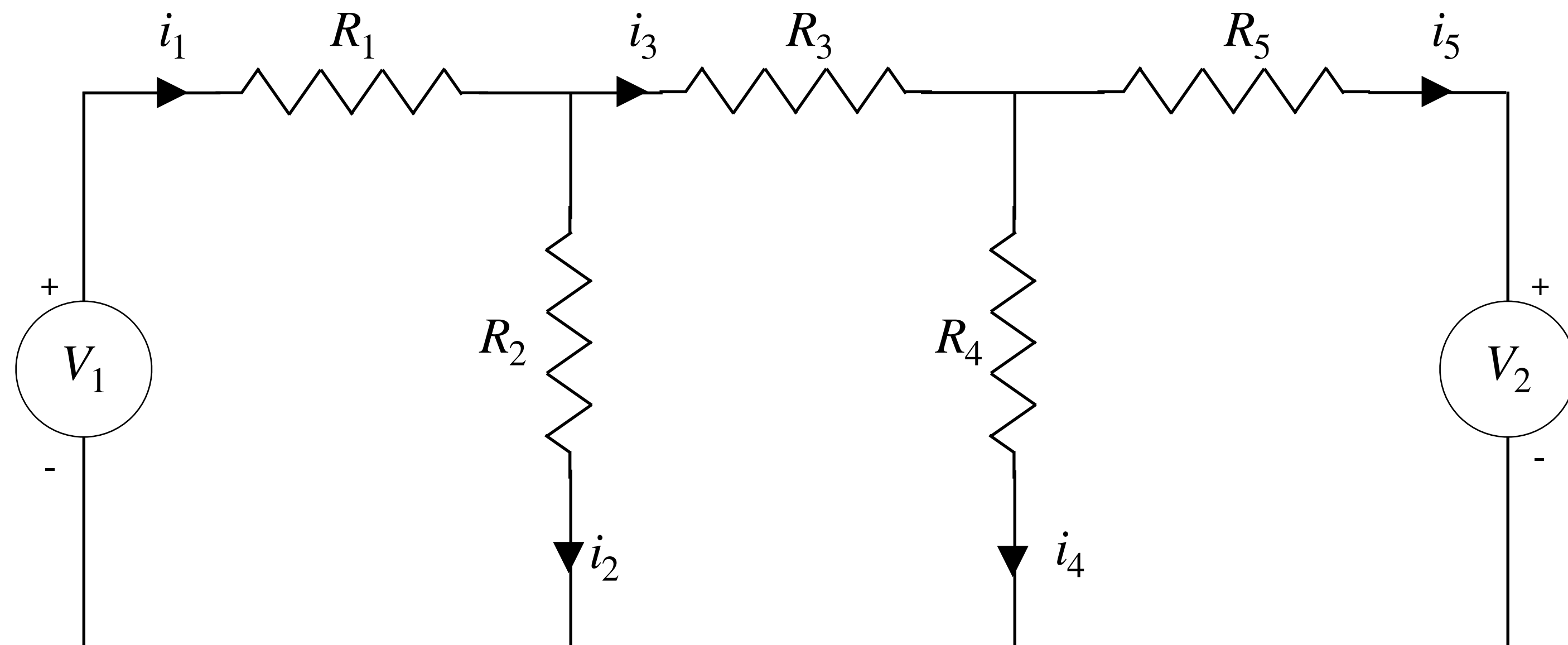
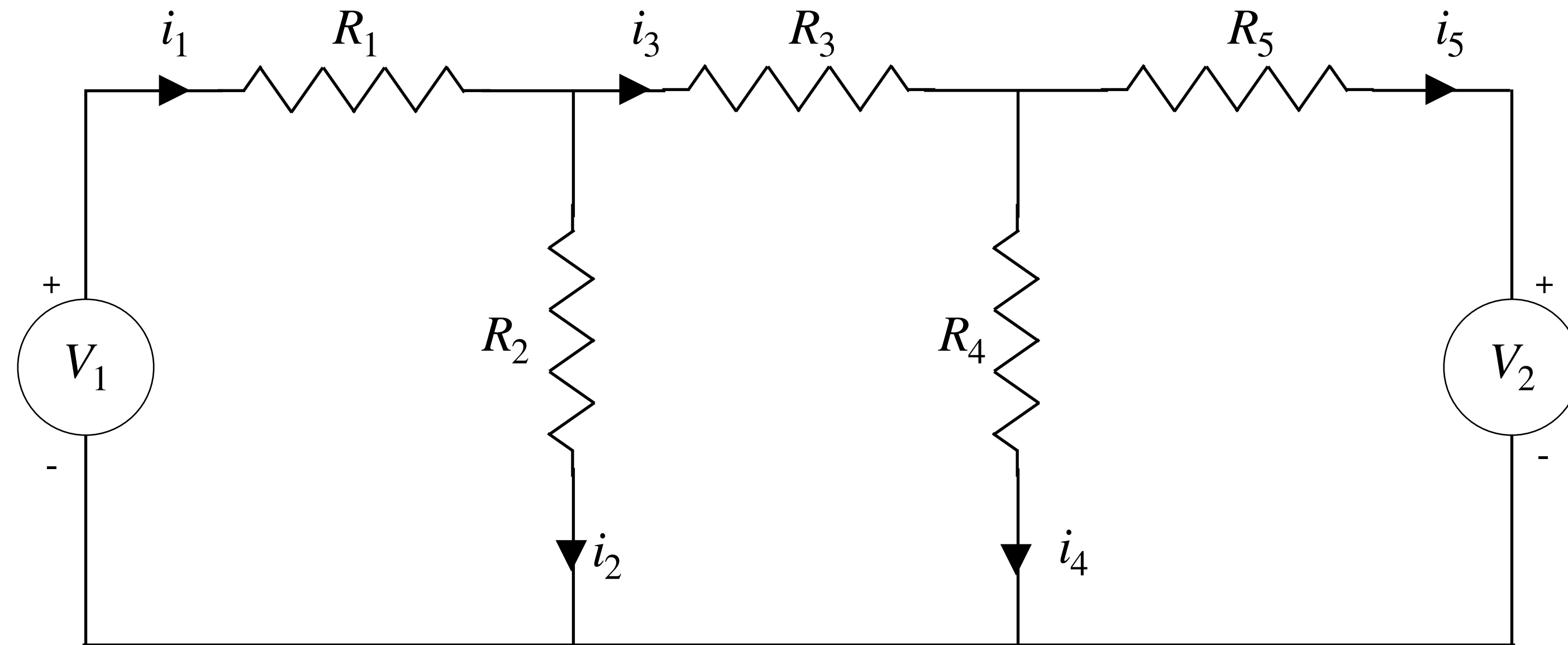


1.4 Solving Linear and Nonlinear equations

Example L05.1: Solve the circuit below for currents $i_1, i_2, \dots, i_5$. You are given that $R_1=5, R_2=150, R_3=100, R_4=250$ and $R_5=200$ kilo ohms. $V_1=100$ and $V_2=50$ volts.





Applying Kirchoff's voltage law gives

$$R_1 i_1 + R_2 i_2 = V_1$$

$$-R_2 i_2 + R_3 i_3 + R_4 i_4 = 0$$

$$R_4 i_4 - R_5 i_5 = V_2$$

We also know that

$$i_1 - i_2 - i_3 = 0$$

$$i_3 - i_4 - i_5 = 0$$

Applying Kirchoff's voltage law gives

$$\begin{aligned} R_1 i_1 + R_2 i_2 &= V_1 \\ -R_2 i_2 + R_3 i_3 + R_4 i_4 &= 0 \\ R_4 i_4 - R_5 i_5 &= V_2 \end{aligned}$$

We also know that

$$\begin{aligned} i_1 - i_2 - i_3 &= 0 \\ i_3 - i_4 - i_5 &= 0 \end{aligned}$$

Hence, in order to find the currents, $i_1, i_2, \dots, i_5$, you need to solve

$$\begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix} \begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix} = \begin{Bmatrix} V_1 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

$$[A] \{X\} = \{C\}$$

where

$$[A] = \begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix}, \quad \{X\} = \begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix}, \quad \{C\} = \begin{Bmatrix} V_1 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

$R_1=5$
 $R_2=150$
 $R_3=100$
 $R_4=250$
 $R_5=200$

$V_1=100.0$
 $V_2=50.0$

$A = \begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix}$

$C_1 = [V_1, 0, V_2, 0, 0]$ #C1 is a (5x1) column vector

$i = A \backslash C_1$

Alternatively

```

# Define the linear problem
prob = LinearProblem(A,C1)

# Solve the problem using defaults
@time sol = solve(prob)
i=sol.u
  
```

$$[A] = \begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix},$$

$$\{C\} = \begin{Bmatrix} V_1 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

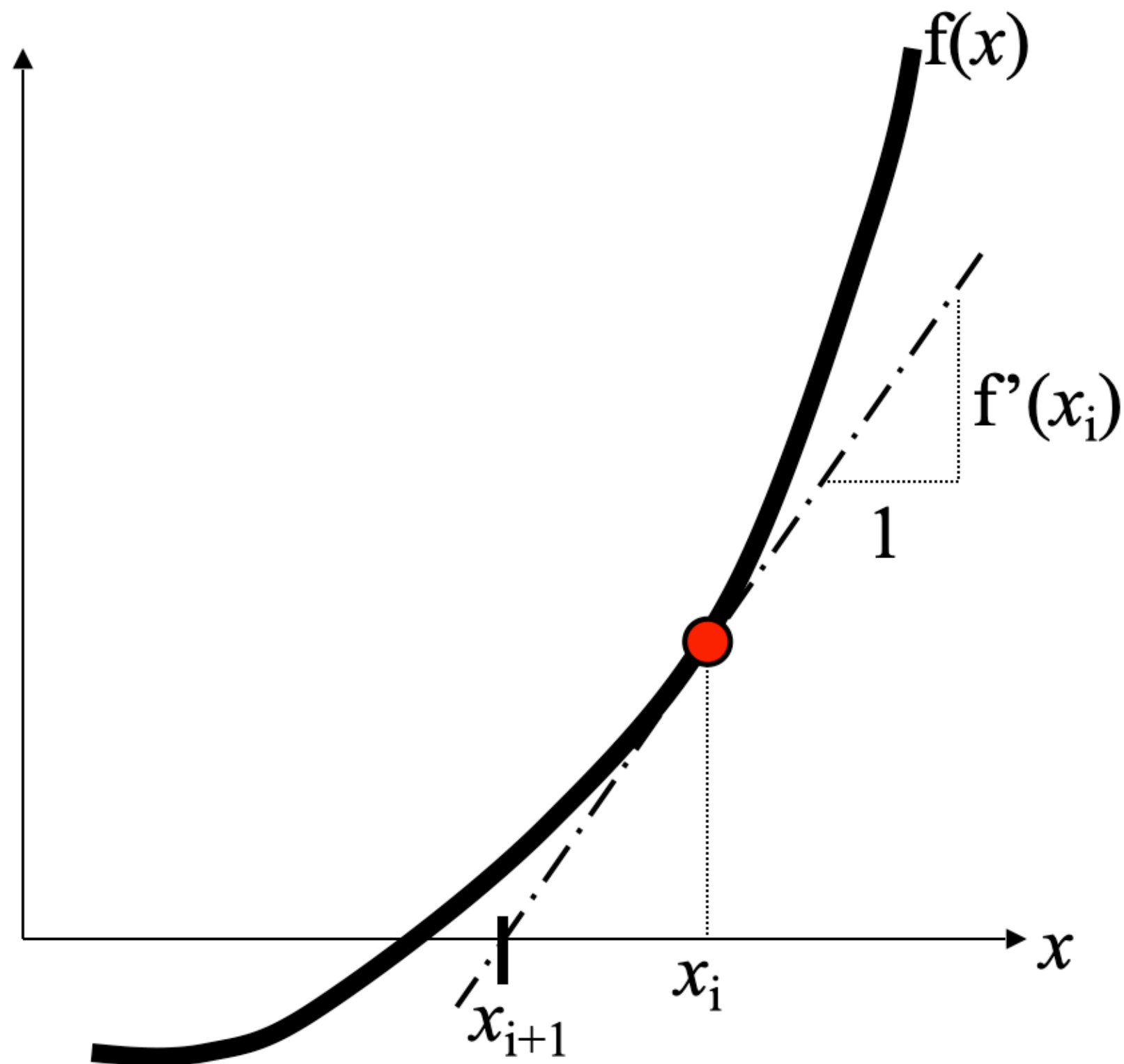
$$\begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix} = \begin{bmatrix} R_1 & R_2 & 0 & 0 & 0 \\ 0 & -R_2 & R_3 & R_4 & 0 \\ 0 & 0 & 0 & R_4 & -R_5 \\ 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{bmatrix}^{-1} \begin{Bmatrix} V_1 \\ 0 \\ V_2 \\ 0 \\ 0 \end{Bmatrix}$$

5-element Vector{Float64}:

$$\begin{Bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{Bmatrix} = \begin{bmatrix} 0.9543568464730301 \\ 0.6348547717842323 \\ 0.31950207468879666 \\ 0.25311203319502074 \\ 0.0663900414937759 \end{bmatrix}$$

Systems of nonlinear equations

Recall Newton-Raphson method for finding roots of the equation $f(x)=0$. This method can be extended to solve a system of nonlinear equations. Let's rederive Newton-Raphson formula (Eq. (3.7) below) using Taylor's series.



From the graph on the left, it is easy to express x_i in terms of $f(x_i)$, $f'(x_i)$ and x_{i+1} .

$$f'(x_i) = \frac{f(x_i)}{x_i - x_{i+1}}$$

$$x_i - x_{i+1} = \frac{f(x_i)}{f'(x_i)}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \quad (3.7)$$

Truncate Taylor series to the 2nd order

$$f(x_{i+1}) = f(x_i) + f'(x_i)(x_{i+1} - x_i)$$

we want to find x_i such that $f(x_{i+1}) = 0$. So setting $f(x_{i+1}) = 0$ in the equation above gives

$$f(x_i) + f'(x_i)(x_{i+1} - x_i) = 0$$

$$x_{i+1} = x_i - f(x_i)/f'(x_i)$$

We would like to minimize

$$A(r) = 2\pi r^2 + \frac{2}{r}$$

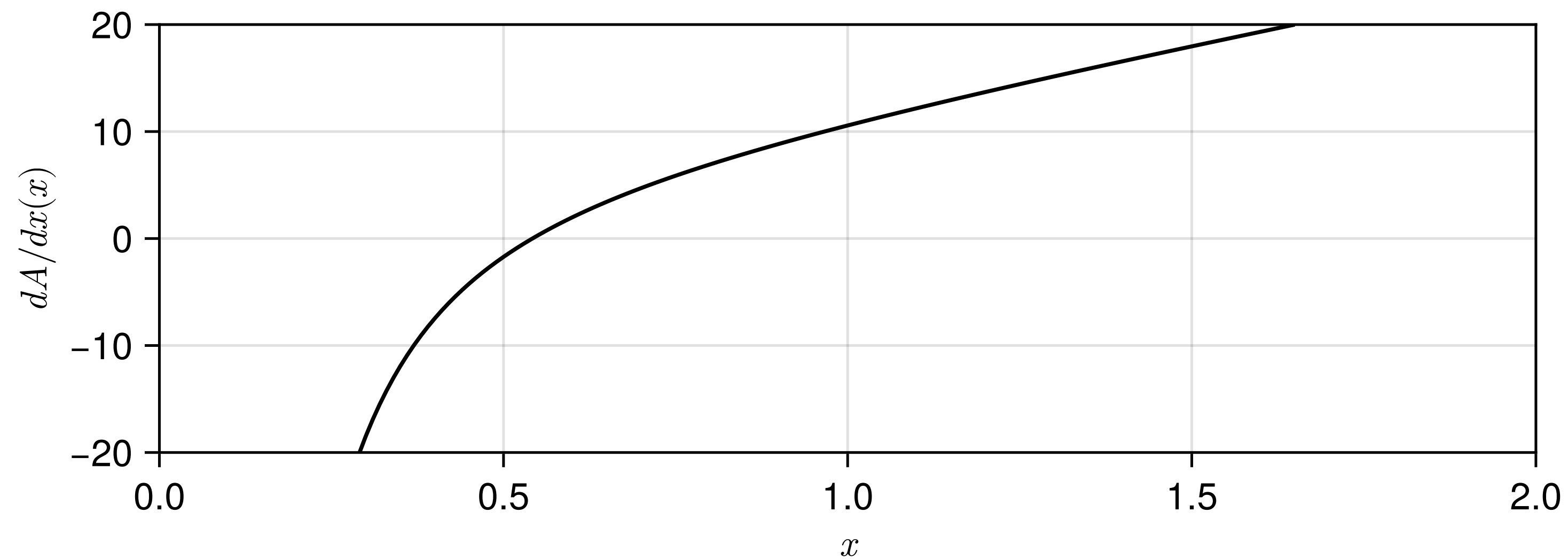
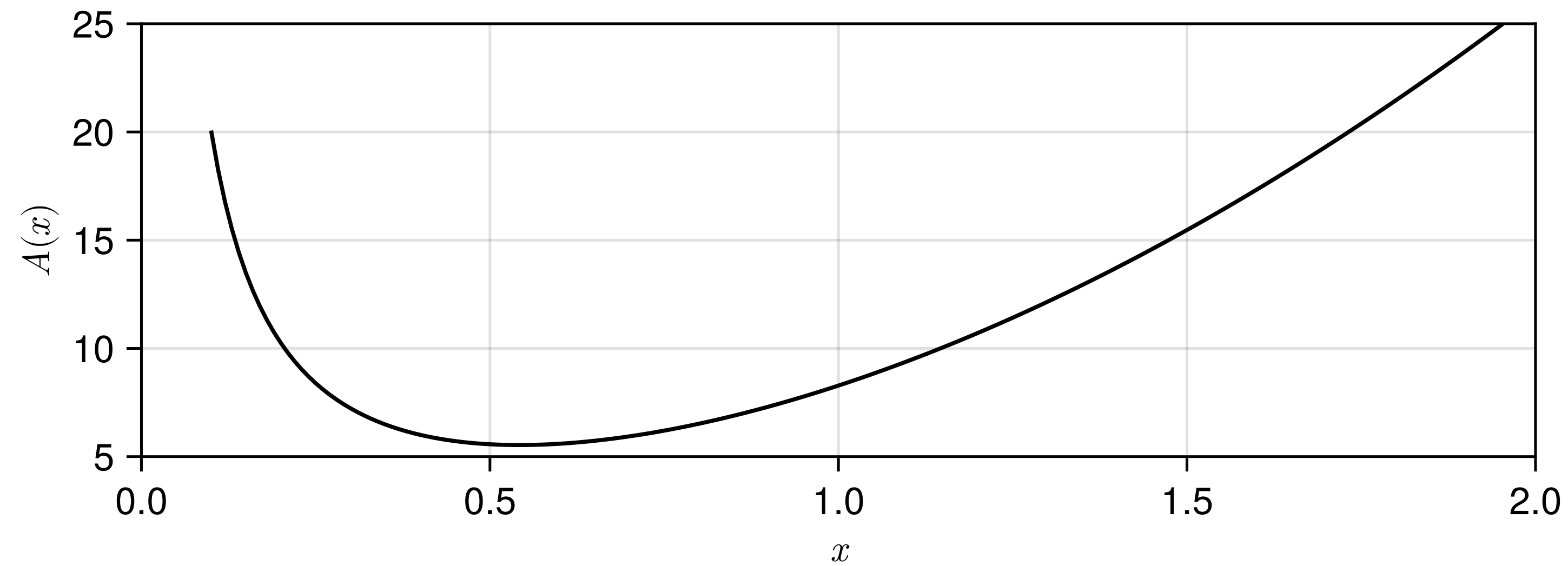
So we want to find r such that

$$dA(r)/dr = 4\pi r - \frac{2}{r^2} = 0$$

Let

$$f(x) = 4\pi x - \frac{2}{x^2}$$

$$\frac{df}{dx}(x) = 4\pi + \frac{4}{x^3}$$



$$f(x) = 4\pi x - \frac{2}{x^2}$$

$$dfdx(x) = 4\pi + \frac{4}{x^3}$$

```
x=1.0 #initial guess
for i=1:10
    x=x-f(x)/dfdx(x)
    println(x)
end
```

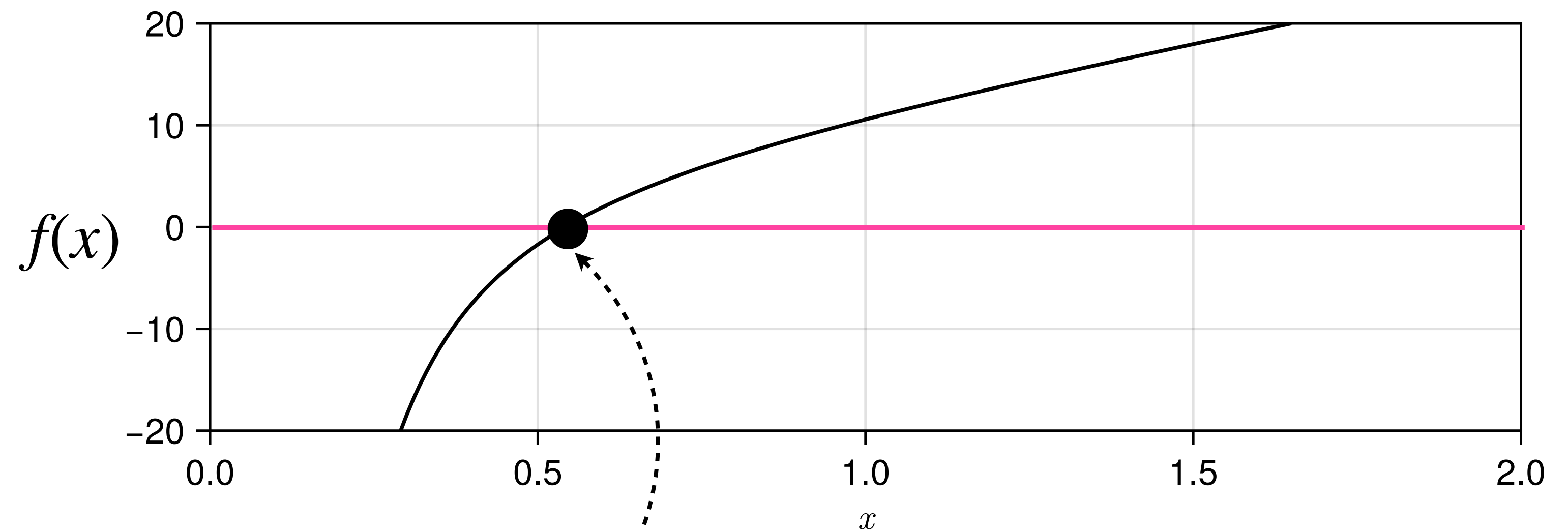
```
0.3621795105078358
0.4727153376004709
0.5323948682518493
0.5417564911696957
0.5419260170637374
0.5419260701392838
0.5419260701392891
0.5419260701392891
0.5419260701392891
0.5419260701392891
```

$$f(x) = 4\pi x - \frac{2}{x^2}$$

$$\frac{df}{dx}(x) = 4\pi + \frac{4}{x^3}$$

Newton-Raphson formula

$$x_{i+1} = x_i - f(x_i)/f'(x_i)$$



```
f(x)=4π*x-2/x^2  
dfdx(x)=4π+4/x^3
```

```
x=1.0 #initial guess  
for i=1:10  
    x=x-f(x)/dfdx(x)  
    println(x)  
end
```

Alternatively

```
x0=1.0 #initial guess  
p=[] #no parameters  
prob = NonlinearProblem((x,p)->f(x),x0,p)  
sol = NonlinearSolve.solve(prob)
```


Sometimes, we need to solve a set of (more than 1) nonlinear equations. I will illustrate with the system of two nonlinear equations

$$u(x, y) = 0$$

$$v(x, y) = 0$$

Performing Taylor series expansion gives

$$u(x_{i+1}, y_{i+1}, z_{i+1}) \approx u(x_i, y_i, z_i) + u_x(x_i, y_i, z_i)(x_{i+1} - x_i) \\ + u_y(x_i, y_i, z_i)(y_{i+1} - y_i) = 0$$

$$v(x_{i+1}, y_{i+1}, z_{i+1}) \approx v(x_i, y_i, z_i) + v_x(x_i, y_i, z_i)(x_{i+1} - x_i) \\ + v_y(x_i, y_i, z_i)(y_{i+1} - y_i) = 0$$

where

$$u_x = \frac{\partial u}{\partial x}, u_y = \frac{\partial u}{\partial y},$$

$$v_x = \frac{\partial v}{\partial x}, v_y = \frac{\partial v}{\partial y},$$

We can put the above equations in Matrix form as

$$\begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix} \begin{Bmatrix} x_{i+1} - x_i \\ y_{i+1} - y_i \end{Bmatrix} = - \begin{Bmatrix} u \\ v \end{Bmatrix}$$

So now to find x_{i+1} and y_{i+1} , you need to solve a system of linear equations

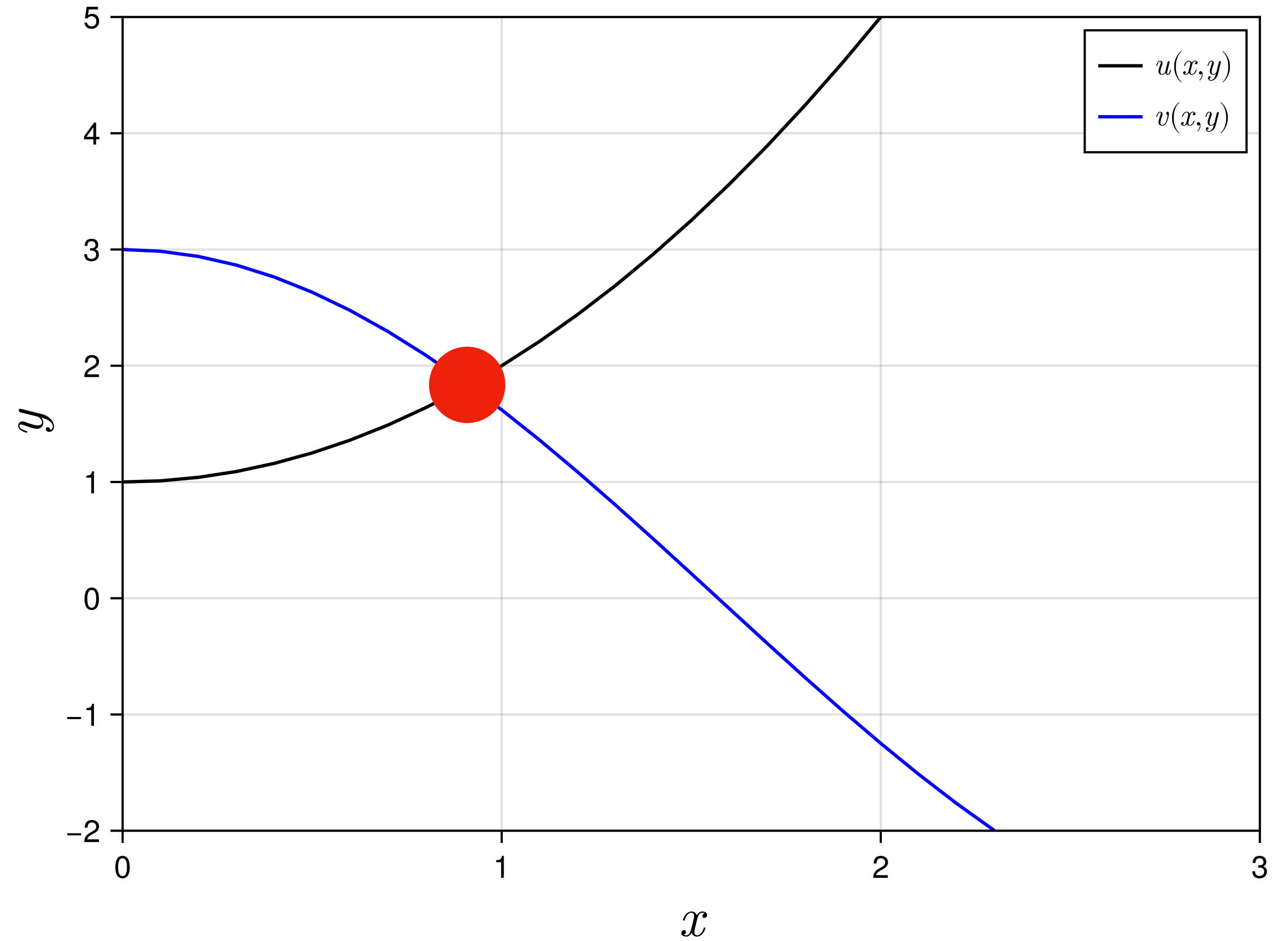
Example 1.4.3

$$y = x^2 + 1$$

$$y = 3 \cos(x)$$

$$u(x, y) = x^2 - y + 1 = 0$$

$$v(x, y) = 3 \cos(x) - y = 0$$




```

function uandv(x,y)
    return [x^2-y+1, 3*cos(x)-y]
end

function Jacobian(x,y)
    return [2*x -1; -3*sin(x) -1]
end

x=y=0.5

for i=1:5
    J=Jacobian(x,y)
    F=uandv(x,y)
    delta=-J\F
    x+=delta[1]
    y+=delta[2]
    println("x=$x, y=$y")
end

```

$$u(x, y) = x^2 - y + 1 = 0$$

$$v(x, y) = 3 \cos(x) - y = 0$$

$$\begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x & -1 \\ -3 \sin(x) & -1 \end{bmatrix}$$

$$\begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix} \begin{Bmatrix} x_{i+1} - x_i \\ y_{i+1} - y_i \end{Bmatrix} = - \begin{Bmatrix} u \\ v \end{Bmatrix}$$

$$\begin{Bmatrix} x_{i+1} - x_i \\ y_{i+1} - y_i \end{Bmatrix} = - \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix}^{-1} \begin{Bmatrix} u \\ v \end{Bmatrix} = \begin{Bmatrix} \delta_1 \\ \delta_2 \end{Bmatrix}$$

$$\begin{Bmatrix} x_{i+1} \\ y_{i+1} \end{Bmatrix} = \begin{Bmatrix} x_i + \delta_1 \\ y_i + \delta_2 \end{Bmatrix}$$

```

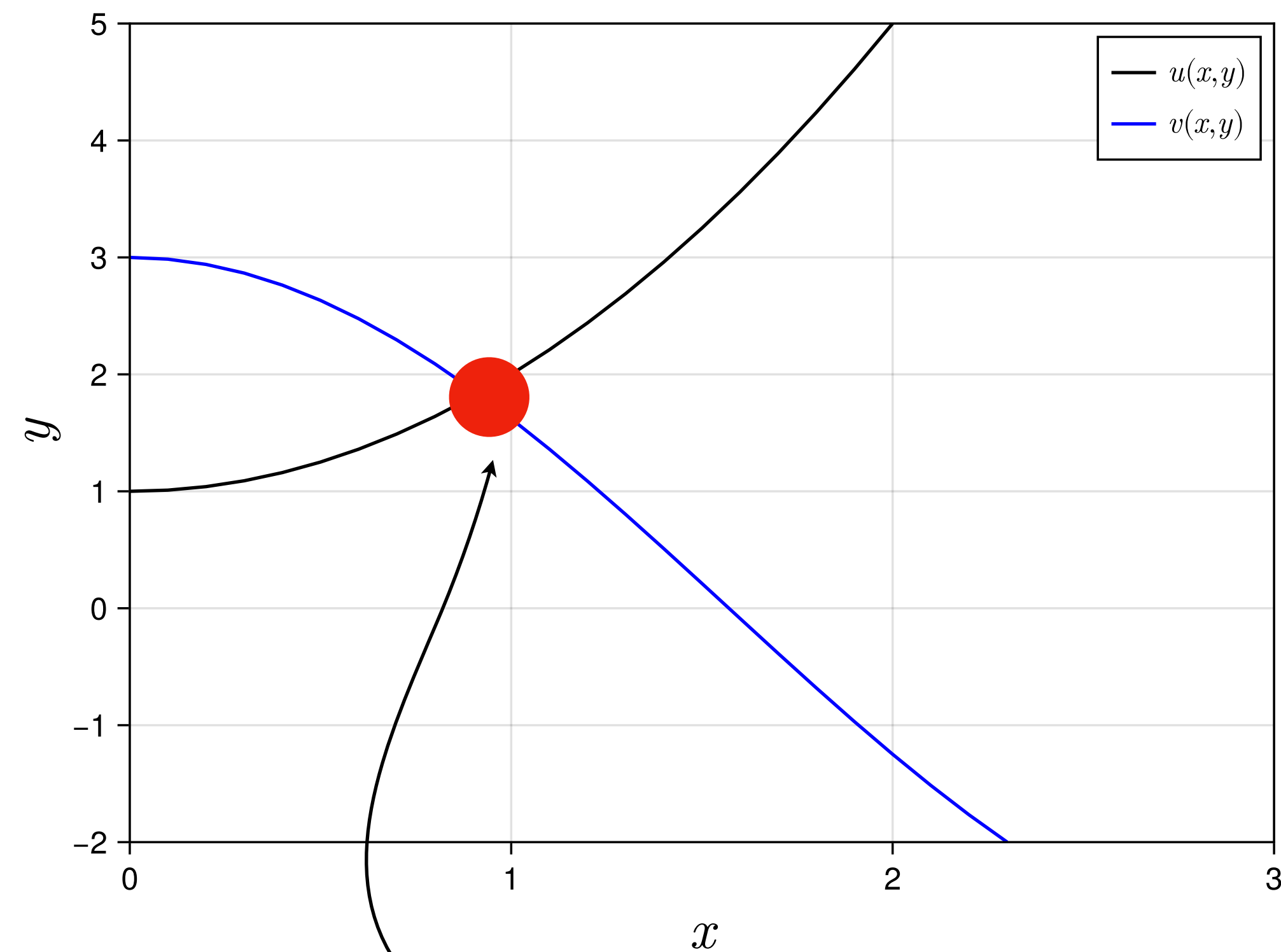
function uandv(x,y)
    return [x^2-y+1,3*cos(x)-y]
end

function Jacobian(x,y)
    return [2*x -1; -3*sin(x) -1]
end

x=y=0.5

for i=1:5
    J=Jacobian(x,y)
    F=uandv(x,y)
    delta=-J\F
    x+=delta[1]
    y+=delta[2]
    println("x=$x, y=$y")
end

```



```

x=1.0671004170338105, y=1.81710041703381
x=0.9220432968528314, y=1.8291222731560397
x=0.9131678157919709, y=1.8337966856342172
x=0.9131320013712529, y=1.8338100506455972
x=0.913132000785916, y=1.83381005085929

```


We would like to solve

$$f(x) = 0$$

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \dots +$$

We truncate Taylor's series at the 2nd order term

$$f(x) = f(x_0) + f'(x_0)(x - x_0)$$

We want to find the value of x such that f(x)=0.
If we set f(x)=0 in the above equation we will get

$$f(x_0) + f'(x_0)(x - x_0) = 0$$

Rearranging and setting $x=x_{i+1}$ and $x_0=x_i$ gives

$$f'(x_i)(x_{i+1} - x_i) = -f(x_i)$$

We can solve for x_{i+1} to get

$$x_{i+1} = x_i - f(x_i)/f'(x_i)$$

We would like to solve

$$f_1(x_1, x_2, x_3) = 0$$

$$f_2(x_1, x_2, x_3) = 0$$

$$f_3(x_1, x_2, x_3) = 0$$

Truncated Taylor's series for a function of 3 variables

$$f_1(x_1, x_2, x_3) = f_1(x_{1,0}, x_{2,0}, x_{3,0}) + \frac{\partial f_1}{\partial x_1}(x_1 - x_{1,0}) + \frac{\partial f_1}{\partial x_2}(x_2 - x_{2,0}) + \frac{\partial f_1}{\partial x_3}(x_3 - x_{3,0})$$

$$f_2(x_1, x_2, x_3) = f_2(x_{1,0}, x_{2,0}, x_{3,0}) + \frac{\partial f_2}{\partial x_1}(x_1 - x_{1,0}) + \frac{\partial f_2}{\partial x_2}(x_2 - x_{2,0}) + \frac{\partial f_2}{\partial x_3}(x_3 - x_{3,0})$$

$$f_3(x_1, x_2, x_3) = f_3(x_{1,0}, x_{2,0}, x_{3,0}) + \frac{\partial f_3}{\partial x_1}(x_1 - x_{1,0}) + \frac{\partial f_3}{\partial x_2}(x_2 - x_{2,0}) + \frac{\partial f_3}{\partial x_3}(x_3 - x_{3,0})$$

All partial derivatives are evaluated at $x_{1,0}$, $x_{2,0}$ and $x_{3,0}$

We want to find x_1 , x_2 and x_3 such that f_1 , f_2 and $f_3=0$.

If we set $f_1=f_2=f_3=0$ on the left hand side of the equations above

$$f_1(x_{1,0}, x_{2,0}, x_{3,0}) + \frac{\partial f_1}{\partial x_1}(x_1 - x_{1,0}) + \frac{\partial f_1}{\partial x_2}(x_2 - x_{2,0}) + \frac{\partial f_1}{\partial x_3}(x_3 - x_{3,0}) = 0$$

$$f_2(x_{1,0}, x_{2,0}, x_{3,0}) + \frac{\partial f_2}{\partial x_1}(x_1 - x_{1,0}) + \frac{\partial f_2}{\partial x_2}(x_2 - x_{2,0}) + \frac{\partial f_2}{\partial x_3}(x_3 - x_{3,0}) = 0$$

$$f_3(x_{1,0}, x_{2,0}, x_{3,0}) + \frac{\partial f_3}{\partial x_1}(x_1 - x_{1,0}) + \frac{\partial f_3}{\partial x_2}(x_2 - x_{2,0}) + \frac{\partial f_3}{\partial x_3}(x_3 - x_{3,0}) = 0$$

Rearranging and setting $x_n=x_{n,i+1}$ and $x_{n,0}=x_{n,i}$

$$\frac{\partial f_1}{\partial x_1}(x_{1,i+1} - x_{1,i}) + \frac{\partial f_1}{\partial x_2}(x_{2,i+1} - x_{2,i}) + \frac{\partial f_1}{\partial x_3}(x_{3,i+1} - x_{3,i}) = -f_1(x_{1,i}, x_{2,i}, x_{3,i})$$

$$\frac{\partial f_2}{\partial x_1}(x_{1,i+1} - x_{1,i}) + \frac{\partial f_2}{\partial x_2}(x_{2,i+1} - x_{2,i}) + \frac{\partial f_2}{\partial x_3}(x_{3,i+1} - x_{3,i}) = -f_2(x_{1,i}, x_{2,i}, x_{3,i})$$

$$\frac{\partial f_3}{\partial x_1}(x_{1,i+1} - x_{1,i}) + \frac{\partial f_3}{\partial x_2}(x_{2,i+1} - x_{2,i}) + \frac{\partial f_3}{\partial x_3}(x_{3,i+1} - x_{3,i}) = -f_3(x_{1,i}, x_{2,i}, x_{3,i})$$

$$\begin{bmatrix} \left(\frac{\partial f_1}{\partial x_1}\right)_i & \left(\frac{\partial f_1}{\partial x_2}\right)_i & \left(\frac{\partial f_1}{\partial x_3}\right)_i \\ \left(\frac{\partial f_2}{\partial x_1}\right)_i & \left(\frac{\partial f_2}{\partial x_2}\right)_i & \left(\frac{\partial f_2}{\partial x_3}\right)_i \\ \left(\frac{\partial f_3}{\partial x_1}\right)_i & \left(\frac{\partial f_3}{\partial x_2}\right)_i & \left(\frac{\partial f_3}{\partial x_3}\right)_i \end{bmatrix} \begin{bmatrix} x_{1,i+1} - x_{1,i} \\ x_{2,i+1} - x_{2,i} \\ x_{3,i+1} - x_{3,i} \end{bmatrix} = \begin{bmatrix} -f_1(x_{1,i}, x_{2,i}, x_{3,i}) \\ -f_2(x_{1,i}, x_{2,i}, x_{3,i}) \\ -f_3(x_{1,i}, x_{2,i}, x_{3,i}) \end{bmatrix}$$

Better guesses
of x_1, x_2 and x_3

functions evaluated at old
guesses of x_1, x_2 and x_3

Old guesses
of x_1, x_2 and x_3

Jacobian

$$\begin{bmatrix} \left(\frac{\partial f_1}{\partial x_1}\right)_i & \left(\frac{\partial f_1}{\partial x_2}\right)_i & \left(\frac{\partial f_1}{\partial x_3}\right)_i \\ \left(\frac{\partial f_2}{\partial x_1}\right)_i & \left(\frac{\partial f_2}{\partial x_2}\right)_i & \left(\frac{\partial f_2}{\partial x_3}\right)_i \\ \left(\frac{\partial f_3}{\partial x_1}\right)_i & \left(\frac{\partial f_3}{\partial x_2}\right)_i & \left(\frac{\partial f_3}{\partial x_3}\right)_i \end{bmatrix}
 \begin{bmatrix} x_{1,i+1} - x_{1,i} \\ x_{2,i+1} - x_{2,i} \\ x_{3,i+1} - x_{3,i} \end{bmatrix} = \begin{bmatrix} -f_1(x_{1,i}, x_{2,i}, x_{3,i}) \\ -f_2(x_{1,i}, x_{2,i}, x_{3,i}) \\ -f_3(x_{1,i}, x_{2,i}, x_{3,i}) \end{bmatrix} \quad (7.8)$$

$$[A] \{X\} = \{C\}$$

So we need to solve a linear system of equation to get a better guess,
and then iterate

Newton's Method for nonlinear equations

Hence, to solve the original system of equations, do the following

1. Calculate all the required derivatives.
2. Guess the (initial) values for $x_{1,i}$, $x_{2,i}$ and $x_{3,i}$.
3. Solve Eq. (7.8) to obtain $(x_{1,i+1}-x_{1,i})$, $(x_{2,i+1}-x_{2,i})$ and $(x_{3,i+1}-x_{3,i})$ You can use Gauss elimination, LU decomposition, Gauss-Seidel etc.
4. Calculate $x_{1,i+1}$, $x_{2,i+1}$, $x_{3,i+1}$.
5. Repeat steps 2, 3 and 4 until converged solution obtained.

This method is a more general version of the Newton-Raphson method taught in Lecture L3.3.